Supporting Document
Mandatory Technical Document

PP-Module for VVoIP
Version: 2.0
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National Information Assurance Partnership

Foreword

This is a Supporting Document (SD), intended to complement the Common Criteria version 3 and the associated Common Evaluation Methodology for Information Technology Security Evaluation.

SDs may be “Guidance Documents”, that highlight specific approaches and application of the standard to areas where no mutual recognition of its application is required, and as such, are not of normative nature, or “Mandatory Technical Documents”, whose application is mandatory for evaluations whose scope is covered by that of the SD. The usage of the latter class is not only mandatory, but certificates issued as a result of their application are recognized under the CCRA.

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Document history:

Version	Date	Comment
1.0	2020-10-28	Initial publication
2.0	2026-03-17	CC:2022 updates, incorporation of TDs

General Purpose:
The purpose of this SD is to define evaluation methods for the functional behavior of VVoIP products.

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1 Introduction

1.1 Technology Area and Scope of Supporting Document

The scope of the PP-Module for VVoIP is to describe the security functionality of VVoIP products in terms of [CC] and to define functional and assurance requirements for them. The PP-Module is intended for use with the following Base-PPs:

- Network Device, version 4.0.
- Application Software, version 2.0.

This SD is mandatory for evaluations of TOEs that claim conformance to a PP-Configuration that includes the PP-Module for :

- VVoIP, Version 2.0

As such it defines Evaluation Activities for the functionality described in the PP-Module as well as any impacts to the Evaluation Activities to the Base-PP(s) it modifies.

Although Evaluation Activities are defined mainly for the evaluators to follow, in general they also help developers to prepare for evaluation by identifying specific requirements for their TOE. The specific requirements in Evaluation Activities may in some cases clarify the meaning of Security Functional Requirements (SFR), and may identify particular requirements for the content of Security Targets (ST) (especially the TOE Summary Specification), user guidance documentation, and possibly supplementary information (e.g. for entropy analysis or cryptographic key management architecture).

1.2 Structure of the Document

Evaluation Activities can be defined for both SFRs and Security Assurance Requirements (SAR), which are themselves defined in separate sections of the SD.

If any Evaluation Activity cannot be successfully completed in an evaluation, then the overall verdict for the evaluation is a 'fail'. In rare cases there may be acceptable reasons why an Evaluation Activity may be modified or deemed not applicable for a particular TOE, but this must be approved by the Certification Body for the evaluation.

In general, if all Evaluation Activities (for both SFRs and SARs) are successfully completed in an evaluation then it would be expected that the overall verdict for the evaluation is a ‘pass’. To reach a ‘fail’ verdict when the Evaluation Activities have been successfully completed would require a specific justification from the evaluator as to why the Evaluation Activities were not sufficient for that TOE.

Similarly, at the more granular level of assurance components, if the Evaluation Activities for an assurance component and all of its related SFR Evaluation Activities are successfully completed in an evaluation then it would be expected that the verdict for the assurance component is a ‘pass’. To reach a ‘fail’ verdict for the assurance component when these Evaluation Activities have been successfully completed would require a specific justification from the evaluator as to why the Evaluation Activities were not sufficient for that TOE.

1.3 Terms

The following sections list Common Criteria and technology terms used in this document.

1.3.1 Common Criteria Terms

Assurance	Grounds for confidence that a TOE meets the SFRs [CC].
Base Protection Profile (Base-PP)	Protection Profile used as a basis to build a PP-Configuration.
Collaborative Protection Profile (cPP)	A Protection Profile developed by international technical communities and approved by multiple schemes.
Common Criteria (CC)	Common Criteria for Information Technology Security Evaluation (International Standard ISO/IEC 15408).

Common Criteria Testing Laboratory	Within the context of the Common Criteria Evaluation and Validation Scheme (CCEVS), an IT security evaluation facility accredited by the National Voluntary Laboratory Accreditation Program (NVLAP) and approved by the NIAP Validation Body to conduct Common Criteria-based evaluations.
Common Evaluation Methodology (CEM)	Common Evaluation Methodology for Information Technology Security Evaluation.
Direct Rationale	A type of Protection Profile, PP-Module, or Security Target in which the security problem definition (SPD) elements are mapped directly to the SFRs and possibly to the security objectives for the operational environment. There are no security objectives for the TOE.
Distributed TOE	A TOE composed of multiple components operating as a logical whole.
Functional Package (FP)	A document that collects SFRs for a particular protocol, technology, or functionality.
Operational Environment (OE)	Hardware and software that are outside the TOE boundary that support the TOE functionality and security policy.
Protection Profile (PP)	An implementation-independent set of security requirements for a category of products.
Protection Profile Configuration (PP-Configuration)	A comprehensive set of security requirements for a product type that consists of at least one Base-PP and at least one PP-Module.
Protection Profile Module (PP-Module)	An implementation-independent statement of security needs for a TOE type complementary to one or more Base-PPs.
Security Assurance Requirement (SAR)	A requirement to assure the security of the TOE.
Security Functional Requirement (SFR)	A requirement for security enforcement by the TOE.
Security Target (ST)	A set of implementation-dependent security requirements for a specific product.
Target of Evaluation (TOE)	The product under evaluation.
TOE Security Functionality (TSF)	The security functionality of the product under evaluation.
TOE Summary Specification (TSS)	A description of how a TOE satisfies the SFRs in an ST.

1.3.2 Technical Terms

Basic telephony functions	The basic functions a VVoIP phone must provide – pick up, dial tone, dial, talk.
Call control	The packets exchanged between devices to establish, maintain, and tear down a telephony call.

Client	A VVoIP endpoint.
Enterprise Session Controller (ESC)	A VVoIP call control server for VVoIP devices.
H.323	A communications protocol defined by ITU-T that is used for creating, modifying, and terminating multimedia sessions with multiple participants.
Peer-to-Peer (P2P)	A VVoIP architecture that forces each peer to be a server and client at the same time.
SDES-SRTP	A method of key negotiation for SRTP.
Secure Real-Time Transport Protocol (SRTP)	A protocol that is used to provide multimedia (voice/video) streaming services with added security of encryption, message authentication and integrity, and replay protection.
Sensitive Data	Call control/signal data, media data, and audit/management data that must be protected while in transit to prevent unauthorized disclosure.
Session Initiation Protocol (SIP)	A communications protocol defined by IETF that is used for creating, modifying, and terminating multimedia sessions with multiple participants.
Software/firmware update delivery channel	A trusted channel between a client and file server (which may be the same server as the call control server) for secure file download of software/firmware updates.
State	A configuration of a VVoIP endpoint based on how the user is currently using or not using the endpoint. Examples include: • Hook state: whether the endpoint is active (off-hook) or inactive (on-hook) • Mute state: the endpoint is active but is deliberately not transmitting data • Hold state: the endpoint is active but is deliberately not transmitting or receiving data
Streaming media	The voice/video exchanged between VVoIP endpoints.
VVoIP Call Control Server	A VVoIP infrastructure device that performs call control functions between a client and other VVoIP endpoints; this may be either a dedicated device such as an ESC or a VVoIP device itself when using P2P.

2 Evaluation Activities for SFRs

The EAs presented in this section capture the actions the evaluator performs to address technology specific aspects covering specific SARs (e.g. ASE_TSS.1, ADV_FSP.1, AGD_OPE.1, and ATE_IND.1) – this is in addition to the CEM workunits that are performed in Section 3 Evaluation Activities for SARs.

Regarding design descriptions (designated by the subsections labeled TSS, as well as any required supplementary material that may be treated as proprietary), the evaluator must ensure there is specific information that satisfies the EA. For findings regarding the TSS section, the evaluator's verdicts will be associated with the CEM workunit ASE_TSS.1-1. Evaluator verdicts associated with the supplementary evidence will also be associated with ASE_TSS.1-1, since the requirement to provide such evidence is specified in ASE in the PP.

For ensuring the guidance documentation provides sufficient information for the administrators/users as it pertains to SFRs, the evaluator's verdicts will be associated with CEM workunits ADV_FSP.1-7, AGD_OPE.1-4, and AGD_OPE.1-5.

Finally, the subsection labeled Tests is where the authors have determined that testing of the product in the context of the associated SFR is necessary. While the evaluator is expected to develop tests, there may be instances where it is more practical for the developer to construct tests, or where the developer may have existing tests. Therefore, it is acceptable for the evaluator to witness developer-generated tests in lieu of executing the tests. In this case, the evaluator must ensure the developer's tests are executing both in the manner declared by the developer and as mandated by the EA. The CEM workunits that are associated with the EAs specified in this section are: ATE_IND.1-3, ATE_IND.1-4, ATE_IND.1-5, ATE_IND.1-6, and ATE_IND.1-7.

2.1 Network Device

The EAs defined in this section are only applicable in cases where the TOE claims conformance to a PP-Configuration that includes the NDcPP.

2.1.1 Modified SFRs

The PP-Module does not modify any requirements when the NDcPP is the base.

2.2 Application Software

The EAs defined in this section are only applicable in cases where the TOE claims conformance to a PP-Configuration that includes the APP PP.

2.2.1 Modified SFRs

2.2.1.1 Protection of the TSF (FPT)

2.2.1.2 Trusted Path/Channel (FTP)

2.3 TOE SFR Evaluation Activities

2.3.1 Communications (FCO)

FCO_VOC_EXT.1 Fixed-Rate Vocoder

FCO_VOC_EXT.1
TSS
The evaluator shall verify that the TSS specifies each vocoder used. The evaluator shall then examine the specification for each vocoder in order to verify that no variable rate vocoders are claimed by the TSF.
Guidance
There are no additional Guidance evaluation activities for this component.
Tests
The evaluator shall set up a test environment that contains the TOE, a call control server (which may be the TOE itself), a network switch, a packet capture tool, and a second VVoIP endpoint. The evaluator shall then perform the following test:

- Test FCO_VOC_EXT.1:1:

- **Step 1:** The evaluator shall ensure the TOE and the second VVoIP endpoint are registered to a call control server or are using P2P.
- **Step 2:** .The evaluator shall use the TOE to dial the second VVoIP endpoint to establish a call and verify the call is established by holding a voice conversation.
- **Step 3:** The evaluator shall review the captured traffic to verify that a fixed rate vocoder is used.

If multiple vocoders are supported, the evaluator shall reconfigure the TOE to use each individual vocoder and repeat steps 1-3 for each vocoder.

2.3.2 User Data Protection (FDP)

FDP_IFC.1 Subset Information Flow Control

FDP_IFC.1
TSS
The evaluator shall verify that the TSS describes how streaming media is not transmitted when not in a streaming media state.
Guidance
There are no additional Guidance evaluation activities for this component.
Tests
Testing of this component is performed through evaluation of FDP_IFF.1.

FDP_IFF.1 Simple Security Attributes

FDP_IFF.1
TSS
The evaluator shall verify that the TSS describes the TOE's enforcement of the media transmission policy and describes the conditions that are necessary for the TSF to transmit voice/video data to the operational environment.
Guidance
There are no additional Guidance evaluation activities for this component.
Tests
The evaluator shall set up a test environment that contains the TOE, a call control server (which may be the TOE itself), a network switch, a traffic capture tool, and a second VVoIP endpoint. The evaluator shall then perform the following tests:

- Test FDP_IFF.1:1: (Conditional - physical TOE)
  - **Step 1:** The evaluator shall place the TOE into the on-hook state without registering it to the call control server or using P2P. The evaluator shall use the traffic capture tool to verify that the TOE does not transmit any streaming media traffic.
  - **Step 2:** The evaluator shall place the TOE into the off-hook state and use the traffic capture tool to verify that the TOE does not transmit any streaming media traffic.
- Test FDP_IFF.1:2: (Conditional - software application TOE)
  - **Step 1:** The evaluator shall execute the VVoIP application without registering it to the call control server or using P2P. The evaluator shall use the traffic capture tool to verify that the TOE does not transmit any streaming media traffic.
  - **Step 2:** The evaluator shall place the TOE into the off-hook state (e.g., by performing a call without specifying any destination data). The evaluator shall use the traffic capture tool to verify that the TOE does not transmit any streaming media traffic.
- Test FDP_IFF.1:3:
  - **Step 1:** The evaluator shall ensure the TOE is using P2P or register the TOE with the call control server and verify that the TOE is registered by checking the call control server screen with current TOE connections and by viewing the call control path traffic using the traffic capture tool.
  - **Step 2:** The evaluator shall place the TOE into the on-hook state and verify using the traffic capture tool that no streaming media traffic is transmitted by the TOE.

    - **Step 3:** The evaluator shall place the TOE into the off-hook state. The evaluator shall then verify that the TOE continues to not transmit streaming media traffic.
- Test FDP_IFF.1:4:
    - **Step 1:** The evaluator shall ensure the TOE is using P2P or shall register the TOE with the call control server and verify that it is registered by checking the call control server with current connections and using the traffic capture tool to verify the call control path traffic.
    - **Step 2:** The evaluator shall ensure the TOE is using P2P or shall register the second VVoIP endpoint with the call control server and verify that it has been registered by checking the call control server with current connections and using the traffic capture tool to verify the call control path traffic.
    - **Step 3:** The evaluator shall use the TOE to dial the second VVoIP endpoint and connect a call. The evaluator shall verify that the connection is made by having a voice/video conversation with the endpoint and using the traffic capture tool to verify that a steady stream of traffic is being transmitted between the two endpoints over the media channel.
    - **Step 4:** The evaluator shall use the TOE to put the call on mute and verify that no traffic is transmitted from the TOE over the media channel to the second VVoIP endpoint. If the TOE is registered to a call control server, the evaluator shall also verify that a mute control message is sent to the call control server and it responds.
    - **Step 5:** The evaluator shall use the TOE to take the call off mute and verify that the streaming media traffic between the TOE and the second VVoIP endpoint is resumed.
- Test FDP_IFF.1:5:
    - **Step 1:** The evaluator shall ensure the TOE is using P2P or register the TOE to the call control server and place it in the on-hook state.
    - **Step 2:** The evaluator shall use a fuzzing tool to attempt to connect to the TOE on the full range of TCP ports used by the TSF. All ports used by the TOE should be closed except for the port that is used to communicate with the ESC.
- Test FDP_IFF.1:6:
    - **Step 1:** The evaluator shall ensure the TOE and the second VVoIP endpoint are registered to a call control server or are using P2P.
    - **Step 2:** The evaluator shall place the TOE in the on-hook state.
    - **Step 3:** The evaluator shall use a fuzzing tool to attempt to connect to the TOE on the full range of UDP ports used by the TSF. All ports used by the TOE should be closed.
    - **Step 4:** The evaluator shall place a call to the second VVoIP endpoint and verify the call is established. The evaluator shall capture the traffic to determine the port used by the TOE to carry the media traffic.
    - **Step 5:** The evaluator shall hang up the call and verify that the TOE has returned to the on-hook state.
    - **Step 6:** The evaluator shall perform fuzzing activities to verify that the port used to carry media traffic in step 4 has been closed.
- Test FDP_IFF.1:7: (Conditional - "The TOE is not in the hold state" is selected in FDP_IFF.1.2)
    - **Step 1:** The evaluator shall use the TOE to put the call on hold and verify that no streaming media traffic is transmitted from the TOE over the media channel. If the TOE is registered to a call control server, the evaluator shall also verify that the VVoIP endpoint on-hold call control is sent to the call control server and it responds.
    - **Step 2:** The evaluator shall use the TOE to take the call off hold and verify that the streaming media traffic between the TOE and the second VVoIP endpoint is resumed.

2.3.3 Security Management (FMT)

FMT_SMF.1/VVoIP Specification of Management Functions (VVoIP Communications)

FMT_SMF.1/VVoIP
TSS
The evaluator shall verify that the TSS provides a description of the TOE initial configuration and describes the ability of the TSF to manage the functions that are defined in the SFR, including how each function is managed (e.g., manually configured, applied via downloaded configuration file).

Guidance
The evaluator shall verify that the operational guidance provides instructions on configuring any functionality specifically related to VVoIP.
Tests
The evaluator shall perform the following tests, depending on the selections made:

- Test FMT_SMF.1/VVoIP:1: (Conditional - "register the TOE to an ESC [manually]" is selected)
  - **Step 1:** On the TOE, input IP address, gateway address, and subnet mask.
  - **Step 2:** If the operational environment is deployed in a manner such that the configuration server and ESC are two distinct servers, input the addresses for each; otherwise, input the ESC address.
  - **Step 3:** Save the configuration.
  - **Step 4:** Verify the TOE registers to the ESC.
- Test FMT_SMF.1/VVoIP:2: (Conditional - "register the TOE to an ESC [via DHCP server]" is selected
  - **Step 1:** On the TOE, input the DHCP server address.
  - **Step 2:** Verify by traffic capture that the TOE receives all needed IP addresses.
  - **Step 3:** Verify by examining the IP address on the TOE.
  - **Step 4:** Verify the TOE registers to the ESC.

  Test EAs to verify that the TOE can act as a VVoIP call control server when using P2P (if it is selected) are performed as part of testing for FDP_IFF.1/CALLCONTROL.

  Test EAs to verify the configuration of audit behavior are performed as part of testing for FAU_STG.1 and FAU_STG.5.

  Test EAs to verify the modification of transmission of audit data to an external IT entity are performed as part of testing for FAU_STG.1.

  Test EAs to verify that the idle call termination period can be specified are performed as part of testing for FTA_SSL.3/MEDIA, specifically Test 2 and Test 3.

  Test EAs to verify that the vocoder can be specified are performed as part of testing for FCO_VOC_EXT.1.

2.3.4 TOE Access (FTA)

FTA_SSL.3/MEDIA TSF-Initiated Termination (Media Channel)

FTA_SSL.3/MEDIA
TSS
The evaluator shall verify that the TSS specifies whether idle calls are terminated by default after a certain amount of time or by an administrator-configurable interval.
Guidance
If the idle call timeout period is administrator-configurable, the evaluator shall verify that the operational guidance includes instructions for how to configure this, as well as what the minimum and maximum allowed values are.
Tests
The evaluator shall set up a test environment that contains the TOE, a call control server (which may be the TOE itself), a configuration server (if used to communicate configuration changes to idle timeout period), a network switch, a traffic capture tool, and a second VVoIP endpoint. The evaluator shall then perform the following tests:

- Test FTA_SSL.3/MEDIA:1:
  - **Step 1:** Deploy the TOE in a default configuration (i.e. without any administrative override applied to the idle timeout value).
  - **Step 2:** Ensure the TOE is using P2P or register the TOE with the call control server and verify that it is registered by viewing its status on the ESC and capturing the call control path traffic.
  - **Step 3:** Ensure the second VVoIP endpoint is using P2P or register the second VVoIP endpoint with the call control server and verify that it is registered by viewing its status on the call control server and

capturing the call control path traffic.
    - **Step 4:** Use the TOE to dial the second VVoIP endpoint and establish a call. Verify the call was established by holding a conversation between the two peers and capturing the streaming media traffic that is transmitted between them.
    - **Step 5:** Power down the second VVoIP endpoint while the call is active. Observe that the TOE stops transmitting media after the default period of time specified in the ST.
- Test FTA_SSL.3/MEDIA:2: (Conditional - "an administrator-configurable interval is selected in FTA_SSL.3.1/MEDIA)
    - **Step 1:** Deploy the TOE in a default configuration (i.e. without any administrative override applied to the idle timeout value).
    - **Step 2:** Ensure the TOE is using P2P or register the TOE with the call control server and verify that it is registered by viewing its status on the call control server and capturing the call control path traffic.
    - **Step 3:** Configure the TOE’s idle timeout period for the shortest period of time that is supported.
    - **Step 4:** Ensure the second VVoIP endpoint is using P2P or register the second VVoIP endpoint with the call control server and verify that it is registered by viewing its status on the call control server and capturing the call control path traffic.
    - **Step 5:** Use the TOE to dial the second VVoIP endpoint and establish a call. Verify the call was established by holding a conversation between the two peers and capturing the streaming media traffic that is transmitted between them.
    - **Step 6:** Power down the second VVoIP endpoint while the call is active. Observe that the TOE stops transmitting media after the period of time configured in Step 3.
- Test FTA_SSL.3/MEDIA:3: (Conditional - "an administrator-configurable interval" is selected in FTA_SSL.3.1/MEDIA)

  Repeat Test 2 but in Step 3, configure the idle timeout value to be the longest period of time that is supported as opposed to the shortest.

2.3.5 Trusted Path/Channels (FTP)

FTP_ITC.1/CONTROL Inter-TSF Trusted Channel (Signaling Channel)

FTP_ITC.1/CONTROL
TSS
The evaluator shall verify that the TSS describes the ability of the TOE to use SIP and/or H.323 with TLS.
Guidance
There are no additional Guidance evaluation activities for this component.
Tests
The vendor shall provide to the evaluator application layer configuration settings for all secure communication mechanisms specified by the FTP_ITC.1/CONTROL requirement. This information should be sufficiently detailed to allow the evaluator to determine the application layer timeout settings for each cryptographic protocol. There is no expectation that this information must be recorded in any public-facing document or report.

The evaluator shall perform the following tests:

- Test FTP_ITC.1/CONTROL:1: The evaluators shall ensure that communications using each protocol with a call control server is tested during the course of the evaluation, setting up the connections as described in the guidance documentation and ensuring that communication is successful.
- Test FTP_ITC.1/CONTROL:2: For each protocol that the TOE can initiate as defined in the requirement, the evaluator shall follow the guidance documentation to ensure that in fact the communication channel can be initiated from the TOE.
- Test FTP_ITC.1/CONTROL:3: The evaluator shall ensure, for each communication channel with an authorized IT entity, the channel data is not sent in plaintext.
- Test FTP_ITC.1/CONTROL:4:

Objective: The objective of this test is to ensure that the TOE reacts appropriately to any connection outage or interruption of the route to the external IT entities.

The evaluator shall, for a connection to a call control server using each claimed protocol, physically interrupt the connection to the call control server for the following durations: i) a duration that exceeds the TOE's application layer timeout setting, ii) a duration shorter than the application layer timeout but of sufficient length to interrupt the MAC layer.

The evaluator shall ensure that, when the physical connectivity is restored, communications are appropriately protected and no TSF data is sent in plaintext.

In the case where the TOE is able to detect when the cable is removed from the device, another physical network device (e.g., a core switch) shall be used to interrupt the connection between the TOE and the call control server. The interruption shall not be performed at the virtual node (e.g., virtual switch) and must be physical in nature.

Further EAs are associated with the specific protocols.

FTP_ITC.1/MEDIA Inter-TSF Trusted Channel (Media Channel)

FTP_ITC.1/MEDIA
TSS
The evaluator shall verify that the trusted channel will use SRTP or H.323/H.235 in accordance with the selections made in the SFR.
Guidance
There are no additional Guidance evaluation activities for this component.
Tests
The vendor shall provide to the evaluator application layer configuration settings for all secure communication mechanisms specified by the FTP_ITC.1/MEDIA requirement. This information should be sufficiently detailed to allow the evaluator to determine the application layer timeout settings for each cryptographic protocol. There is no expectation that this information must be recorded in any public-facing document or report.

The evaluator shall perform the following tests:

- Test FTP_ITC.1/MEDIA:1: The evaluators shall ensure that communications using each protocol with another VVoIP endpoint is tested during the course of the evaluation, setting up the connections as described in the guidance documentation and ensuring that communication is successful.
- Test FTP_ITC.1/MEDIA:2: For each protocol that the TOE can initiate as defined in the requirement, the evaluator shall follow the guidance documentation to ensure that in fact the communication channel can be initiated from the TOE.
- Test FTP_ITC.1/MEDIA:3: The evaluator shall ensure, for each communication channel with an authorized IT entity, the channel data is not sent in plaintext.
- Test FTP_ITC.1/MEDIA:4:

  Objective: The objective of this test is to ensure that the TOE reacts appropriately to any connection outage or interruption of the route to the external IT entities.

  The evaluator shall, for a connection to another VVoIP endpoint device using each claimed protocol, physically interrupt the connection to that remote endpoint for the following durations: i) a duration that exceeds the TOE's application layer timeout setting, ii) a duration shorter than the application layer timeout but of sufficient length to interrupt the MAC layer.

  The evaluator shall ensure that, when the physical connectivity is restored, communications are appropriately protected and no TSF data is sent in plaintext.

In the case where the TOE is able to detect when the cable is removed from the device, another physical network device (e.g., a core switch) shall be used to interrupt the connection between the TOE and the call control server. The interruption shall not be performed at the virtual node (e.g., virtual switch) and must be physical in nature.

Further EAs are associated with the specific protocols.

2.4 Evaluation Activities for Optional SFRs

2.4.1 Security Audit(FAU)

FAU_GEN.1/CSADMIN Audit Data Generation (Client-Server Admin Events)

FAU_GEN.1/CSADMIN
TSS
There are no additional TSS evaluation activities for this component.
Guidance
The evaluator shall make a determination of the administrative actions related to TSF data related to configuration changes. The evaluator shall examine the guidance documentation and make a determination of which administrative commands, including subcommands, scripts, and configuration files, are related to the configuration (including enabling or disabling) of the mechanisms implemented in the TOE that are necessary to enforce the requirements specified in the PP-Module. The evaluator shall document the methodology or approach taken to TSF data related to configuration changes. The evaluator may perform this activity as part of the activities associated with ensuring that the corresponding guidance documentation satisfies the requirements related to it.
Tests
For each administrative action identified in FAU_GEN.1.1/CSADMIN, the evaluator shall perform an action either on the TOE or on the operational environment that causes the event to occur. The evaluator shall verify in each case that an auditable event was generated in a format consistent with the guidance documentation and that all audit record details specified in the SFR are present.

Note that the testing here can be accomplished in conjunction with the testing of the security mechanisms directly.

FAU_GEN.1/CSVVOIP Audit Data Generation (Client-Server VVoIP Events)

FAU_GEN.1/CSVVOIP
TSS
There are no additional TSS evaluation activities for this component.
Guidance
The evaluator shall check the guidance documentation and ensure that it provides an example of each auditable event required by the PP-Module (i.e. at least one instance of each auditable event – comprising the mandatory, optional and selection-based SFR sections as applicable – shall be provided from the actual audit record).
Tests

For each administrative action identified in the Auditable Events table in the PP-Module, the evaluator shall perform an action on the TOE that causes the event to occur. The evaluator shall verify in each case that an auditable event was generated in a format consistent with the guidance documentation and that all audit record details specified in the SFR are present.

Note that the testing here can be accomplished in conjunction with the testing of the security mechanisms directly.

2.5 Evaluation Activities for Selection-Based SFRs

2.5.1 Security Audit (FAU)

FAU_GEN.1/P2PADMIN Audit Data Generation (Peer-to-Peer Admin Events)

FAU_GEN.1/P2PADMIN
TSS
There are no additional TSS evaluation activities for this component.
Guidance
The evaluator shall make a determination of the administrative actions related to TSF data related to configuration changes. The evaluator shall examine the guidance documentation and make a determination of which administrative commands, including subcommands, scripts, and configuration files, are related to the configuration (including enabling or disabling) of the mechanisms implemented in the TOE that are necessary to enforce the requirements specified in the PP-Module. The evaluator shall document the methodology or approach taken to TSF data related to configuration changes. The evaluator may perform this activity as part of the activities associated with ensuring that the corresponding guidance documentation satisfies the requirements related to it.
Tests
For each administrative action identified in FAU_GEN.1.1/P2PADMIN, the evaluator shall perform an action either on the TOE or on the operational environment that causes the event to occur. The evaluator shall verify in each case that an auditable event was generated in a format consistent with the guidance documentation and that all audit record details specified in the SFR are present.

Note that the testing here can be accomplished in conjunction with the testing of the security mechanisms directly.

FAU_GEN.1/P2PVVOIP Audit Data Generation (Peer-to-Peer VVoIP Events)

FAU_GEN.1/P2PVVOIP
TSS
There are no additional TSS evaluation activities for this component.
Guidance
The evaluator shall check the guidance documentation and ensure that it provides an example of each auditable event required by the PP-Module (i.e. at least one instance of each auditable event – comprising the mandatory, optional and selection-based SFR sections as applicable – shall be provided from the actual audit record).
Tests

For each administrative action identified in the Auditable Events table in the PP-Module, the evaluator shall perform an action on the TOE that causes the event to occur. The evaluator shall verify in each case that an auditable event was generated in a format consistent with the guidance documentation and that all audit record details specified in the SFR are present.

Note that the testing here can be accomplished in conjunction with the testing of the security mechanisms directly.

2.5.2 Cryptographic Support (FCS)

FCS_COP.1/SRTP Cryptographic Operation (Encryption/Decryption for SRTP

FCS_COP.1/SRTP
TSS
The evaluator shall examine the TSS to determine that it claims the use of GCM. The evaluator shall verify that the TSS notes the GCM key sizes that the TOE supports for each use of GCM. If GCM is supported for other uses, the evaluator shall ensure that the specific key sizes supported for each use of GCM are identified.
Guidance
There are no additional Guidance evaluation activities for this component.

Tests

If the TOE claims the App PP as its Base-PP, the evaluator shall perform the AES-GCM test activities identified for FCS_COP.1/SKC in the Base-PP.

If the TOE claims the NDcPP as its Base-PP, the evaluator shall perform the AES-GCM test activities identified for FCS_COP.1/DataEncryption in the Base-PP.

FCS_SRTP_EXT.1 Secure Real-Time Transport Protocol

FCS_SRTP_EXT.1
TSS
The evaluator shall examine the TSS to verify that it describes how the SRTP session is negotiated for both incoming and outgoing calls. This includes how the keying material is established, as well has how requests to use the NULL algorithm or other unallowed cipher suites are rejected by the TSF.
Guidance
The evaluator shall examine the operational guidance to determine that it includes instructions for how to disable the use of the SRTP NULL algorithm and how to specify the ports to be used for SRTP communications.
Tests

The evaluator shall follow the procedure for initializing their device so that they are ready to receive and place calls. The evaluator shall then both place and receive a call and determine that the traffic sent and received by the TOE is encrypted using SRTP with the specified cipher suite. The evaluator may choose one of the below two options to ensure that the call is being encrypted and to view the cipher suite being used.

Option 1: The evaluator shall configure the SIP server to report whether SRTP is being used, and if so, print the negotiated SRTP cipher suite. The evaluator shall confirm that SRTP was used for the calls and that the correct cipher suite was negotiated.

Option 2: A packet capture tool should be used with the SIP server's private key loaded in. The evaluator shall decrypt the TLS-SIP traffic, view the SDES negotiation, and ensure that the correct cipher suite was negotiated.

Option 3: The evaluator shall inspect the VVoIP endpoint's audit or system log to ensure that the correct cipher suite was negotiated.

Next, the evaluator shall configure the SIP server to only allow the SRTP NULL cipher suite. The evaluator shall attempt to both place and receive a call and confirm that both attempts failed.

2.5.3 User Data Protection (FDP)

FDP_IFC.1/CALLCONTROL Subset Information Flow Control (for Call Control)

FDP_IFC.1/CALLCONTROL
TSS
There are no additional TSS evaluation activities for this component.
Guidance
There are no additional Guidance evaluation activities for this component.
Tests
Testing of this component is performed through evaluation of FDP_IFF.1/CALLCONTROL.

FDP_IFF.1/CALLCONTROL Subset Information Flow Control (for Call Control)

FDP_IFF.1/CALLCONTROL

TSS
The evaluator shall examine the TSS to verify that it describes the call control protocol(s) used by the TOE and any explicit circumstances under which the TSF will and will not transmit call control data as defined in FDP_IFF.1.4/CALLCONTROL and FDP_IFF.1.5/CALLCONTROL. The TSF should only transmit call control data using the protocol(s) and method(s) of endpoint identification as indicated in FDP_IFC.1/CALLCONTROL or if defined any explicit allow/deny rules in FDP_IFF.1.4/CALLCONTROL and FDP_IFF.1.5/CALLCONTROL.
Guidance
If any aspects of the TOE's call control functionality are configurable (such as the specific call control protocol used or the circumstances in which the TSF will or will not transmit call control data), the evaluator shall examine the guidance documentation to verify that instructions for configuring this behavior are provided.
Tests
The evaluator shall perform one or more of the following tests depending on the protocols that the TOE claims to support.

- Test FDP_IFF.1/CALLCONTROL:1: (Conditional - if "SIP" is selected in FDP_IFF.1.1/CALLCONTROL)
  - **Step 1:** Ensure the TOE is configured to use P2P and act as a call control server using the SIP protocol.
  - **Step 2:** Use a traffic capture tool to capture call-signaling packets traversing the TOE.
  - **Step 3:** Place a call to the TOE from another VVoIP endpoint using the SIP call control protocol to the TOE and observe via packet capture that the TSF established a connection between itself and the peer.
  - **Step 4:** Repeat step 3 with a call placed from the TOE to the other VVoIP endpoint.
  - **Step 5:** (Conditional - if the ST claims any additional rules in FDP_IFF.1.4/CALLCONTROL) Implement any rules assigned in FDP_IFF.1.4 and verify that the specified authorized information flows occur.
  - **Step 6:** (Conditional - if the ST claims any additional rules in FDP_IFF.1.5/CALLCONTROL) Implement any rules assigned in FDP_IFF.1.5 and verify that the specified information flows do not occur.
  - **Step 7:** Repeat steps 1-6 for each method of endpoint identification.
- Test FDP_IFF.1/CALLCONTROL:2: (Conditional - if "H.323" is selected in FDP_IFF.1.1/CALLCONTROL)
  - **Step 1:** Ensure the TOE is configured to use P2P and act as a call control server using the H.323 protocol.
  - **Step 2:** Use a traffic capture tool to capture call-signaling packets traversing the TOE.
  - **Step 3:** Place a call to the TOE from another VVoIP endpoint using the H.323 call control protocol to the TOE and observe via packet capture that the TSF established a connection between itself and the peer.
  - **Step 4:** Repeat step 3 with a call placed from the TOE to the other VVoIP endpoint.
  - **Step 5:** (Conditional - if the ST claims any additional rules in FDP_IFF.1.4/CALLCONTROL) Implement any rules assigned in FDP_IFF.1.4/CallControl and verify that the specified authorized information flows occur.
  - **Step 6:** (Conditional - if the ST claims any additional rules in FDP_IFF.1.5/CALLCONTROL) Implement any rules assigned in FDP_IFF.1.5 and verify that the specified information flows do not occur.
  - **Step 7:** Repeat steps 1-6 for each method of endpoint identification.

2.5.4 Protection of the TSF (FPT)

FPT_STM_EXT.1/VVoIP Reliable Time Stamps (VVoIP Communications)

FPT_STM_EXT.1/VVoIP
TSS

The evaluator shall examine the TSS to ensure that it lists each security function that makes use of time, and that it provides a description of how the time is maintained and considered reliable in the context of each of the time

related functions.

If an ESC is selected in FPT_STM_EXT.1.1, the evaluator shall also verify that the TSS describes the ability of the TOE to support NTP synchronization with an ESC.

Guidance
The evaluator shall review the guidance to confirm that it provides instructions for how to synchronize the time. If an ESC is selected in FPT_STM_EXT.1.1, this should include instructions on how to enable NTP synchronization with an ESC.
Tests
(Conditional - "an ESC" is selected in FPT_STM_EXT.1.2/VVoIP)

The evaluator shall use the guidance documentation to configure the NTP client on the TOE and set up a communication path with the ESC. The evaluator will observe that the ESC has set the time to what is expected.

2.6 Evaluation Activities for Objective SFRs

The PP-Module does not define any objective requirements.

2.7 Evaluation Activities for Implementation-dependent SFRs

2.7.1 Security Audit (FAU)

FAU_STG.1 Audit data storage location

FAU_STG.1
TSS

The evaluator shall examine the TSS to ensure it describes the means by which the audit data are transferred to the external audit server, and how the trusted channel is provided.

The evaluator shall examine the TSS to ensure it describes the amount of audit data that are stored locally and whether that data is stored by the TOE or through the TOE’s invocation of a platform auditing mechanism.

The evaluator shall examine the TSS to ensure that it details whether the transmission of audit information to an external IT entity can be done in real-time or periodically. In case the TOE does not perform transmission in real-time the evaluator needs to verify that the TSS provides details about what event stimulates the transmission to be made as well as the possible acceptable frequency for the transfer of audit data.

Guidance

The evaluator shall examine the guidance documentation to ensure it describes how to establish the trusted channel to the audit server, as well as describe any requirements on the audit server (particular audit server protocol, version of the protocol required, etc.), as well as configuration of the TOE needed to communicate with the audit server.

The evaluator shall also examine the guidance documentation to determine that it describes the relationship between the local audit data and the audit data that are sent to the audit log server. For example, when an audit event is generated, is it simultaneously sent to the external server and the local store, or is the local store used as a buffer and “cleared” periodically by sending the data to the audit server.

Tests

Testing of the trusted channel mechanism for audit will be performed as specified in the associated EAs for the particular trusted channel mechanism. The evaluator shall perform the following additional tests for this requirement:

- Test FAU_STG.1:1:

  The evaluator shall establish a session between the TOE and the audit server according to the configuration guidance provided. The evaluator shall then examine the traffic that passes between the audit server and the TOE during several activities of the evaluator's choice designed to generate audit data to be transferred to the audit server. The evaluator shall observe that these data are not able to be viewed in the clear during this transfer, and that they are successfully received by the audit server. The evaluator shall record the particular software (name, version) used on the audit server during testing. The evaluator shall verify that the TOE is capable of transferring audit data to an external audit server automatically without administrator intervention.

  If the ability to modify this behavior is configurable (e.g., if the TOE can be configured to transmit audit data either to an ESC to which it is registered or to an external trusted IT entity that is not an ESC), the evaluator shall repeat this test as necessary to show that changing the available configuration options will result in the TOE performing the expected behavior in each case.

FAU_STG.5 Prevention of audit data loss

FAU_STG.5
TSS

The evaluator shall examine the TSS to ensure it describes what happens when the local audit data store is full; and how these records are protected against unauthorized access.

The evaluator shall examine the TSS to ensure that it details the behavior of the TOE when the storage space for audit data is full. When the option 'overwrite the oldest stored audit records' is selected this description should include an outline of the rule for overwriting audit data. If 'other actions...' are chosen such as sending the new audit data to an external IT entity, then the related behavior of the TOE shall also be detailed in the TSS.

Guidance

The evaluator shall ensure that the guidance documentation describes all possible configuration options for FAU_STG.5 and the resulting behavior of the TOE for each possible configuration. The description of possible configuration options and resulting behavior shall correspond to those described in the TSS.

Tests

- Test FAU_STG.5:1:

  The evaluator shall perform operations that generate audit data and verify that this data is stored locally. The evaluator shall perform operations that generate audit data until the local storage space is exceeded and verifies that the TOE complies with the behavior defined in FAU_STG.5. Depending on the configuration this means that the evaluator has to check the content of the audit data when the audit data is just filled to the maximum and then verifies that:

  - The audit data remains unchanged with every new auditable event that should be tracked but that the audit data is recorded again after the local storage for audit data is cleared (for the option 'ignore audited events' in FAU_STG.5).
  - The oldest audit data is overwritten with every new auditable event that should be tracked according to the specified rule (for the option 'overwrite the oldest stored audit records' in FAU_STG.5).
  - The TOE behaves as specified (for the option 'other action' in FAU_STG.5).

If the behavior of the TOE when the local audit storage space is full is configurable, the evaluator shall repeat this test for each configuration option and observe that the intended result occurs.

If the local storage location of the audit data is configurable, the evaluator shall repeat this test as needed to demonstrate that specifying a new location for storage of audit data results in audit data being stored to the desired location.

3 Evaluation Activities for SARs

The PP-Module does not define any SARs beyond those defined within the base-PP to which it must claim conformance. It is important to note that a TOE that is evaluated against the PP-Module is inherently evaluated against the Base-PP as well. The Base-PP includes a number of Evaluation Activities associated with both SFRs and SARs. Additionally, the PP-Module includes a number of SFR-based Evaluation Activities that similarly refine the SARs of the Base-PPs. The evaluation laboratory will evaluate the TOE against the chosen Base-PP and supplement that evaluation with the necessary SFRs that are taken from the PP-Module.

4 Required Supplementary Information

This Supporting Document has no required supplementary information beyond the ST, operational guidance, and testing.

Appendix A - References

Identifier	Title
[AppPP]	Protection Profile for Application Software, Version 2.0, June 16, 2025
[NDcPP]	collaborative Protection Profile for Network Devices, Version 4.0, December 22, 2025